

AEDA : An Easier Data Augmentation Technique for Text Classification

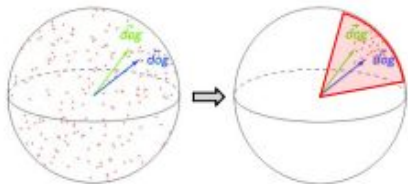
EMNLP 2021 Findings

딥러닝 논문읽기 자연어처리 1팀

발표자 : 김재민

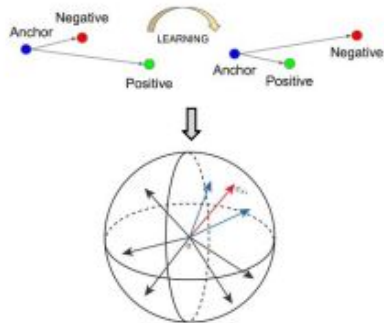
왜 ? AEDA?

Sentence representation learning



- ☑ Universal sentence representation
→ The key to the success of various task
- ☑ Recently, strong potential을 갖고 있는 PLM을 활용
→ 하지만, Anisotropy한 문제를 갖고 있음

Contrastive learning



- ☑ Alignment & Uniformity in semantic representation space
- ☑ Positive pair를 구성하기 위한 연구가 활발
→ 하지만, randomly sample negatives는 거의 없음

Virtual Adversarial Training



- ☑ NLP에서 adversarial perturbation이 embedding layer에서 많이 이뤄짐
→ Text classification, machine translation, NLU task에서 효과적임을 보여줌

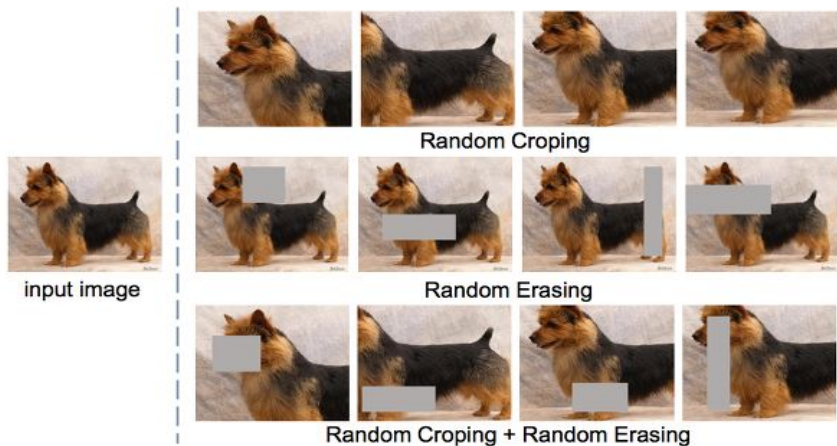


Q&A

1. Background

Data augmentation(DA) ?

IMAGE



출처 : Random Erasing Data Augmentation (<https://arxiv.org/pdf/1708.04896.pdf>)

TEXT

Operation	Sentence
None	A sad, superior human comedy played out on the back roads of life.
SR	A <i>lamentable</i> , superior human comedy played out on the <i>backward</i> road of life.
RI	A sad, superior human comedy played out on <i>funniness</i> the back roads of life.
RS	A sad, superior human comedy played out on <i>roads</i> back <i>the</i> of life.
RD	A sad, superior human out on the roads of life.

Table 1: Sentences generated using EDA. SR: synonym replacement. RI: random insertion. RS: random swap. RD: random deletion.

출처 : EDA: Easy Data Augmentation Techniques for Boosting Performance on Text Classification Tasks (<https://arxiv.org/pdf/1901.11196.pdf>)

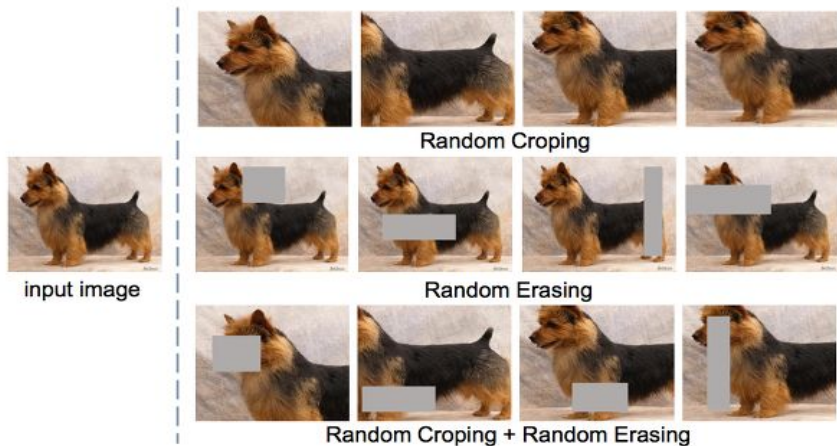
image 와 달리 text에서는 augmentation에 대한 제약을 갖고 있음

최근 많이 활용되는 Back Translation 등이 있음.

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TEXT

Operation	Sentence
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한계점 제시

1. 문장 자체의 정보를 잃을 수 있음
2. input sequence에 따른 label이 변경되어 학습에 misleading 유발

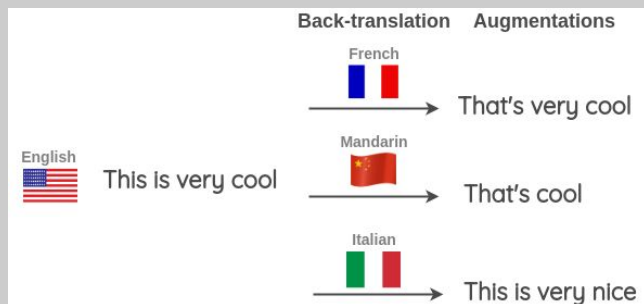
RD: random deletion.

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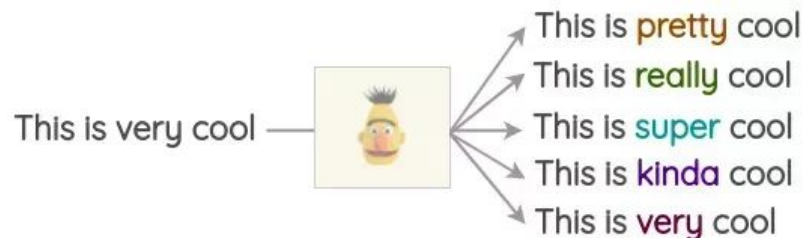
최근 많이 활용되는 Back Translation 등이 있음.

2. Related Work

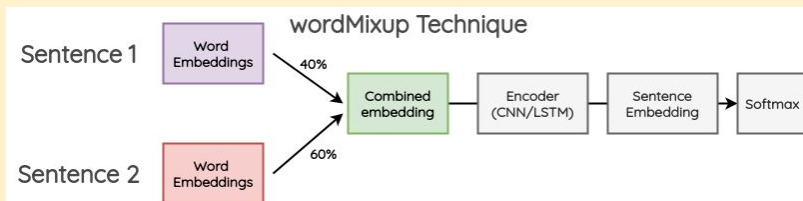
1. Machine Translation



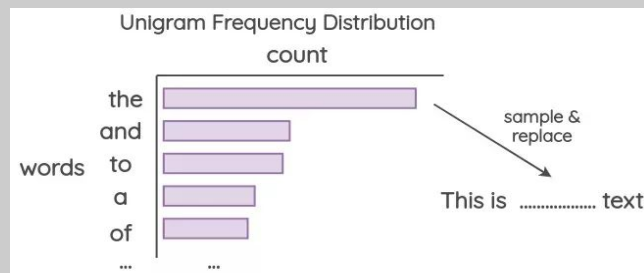
3. LM 을 활용한 Contextual word



2. Mix - Transformer (linearly interpolated)



4. 빈도 기반 단어 대체





Q&A

3. AEDA Augmentation

Original	a sad , superior human comedy played out on the back roads of life .
Aug 1	a sad , superior human comedy played out on the back roads. of life .
Aug 2	a , sad , , superior human , comedy. played . out on the back roads of life .
Aug 3	. a sad ; , superior ! human : comedy , played out ? on the back roads of life .

1. 1~3/1 에 해당하는 **sequence length** 사이의 숫자를 랜덤으로 **position** 지정하여 삽입
2. 선택된 포지션들은 각 6개의 구두점 중 랜덤으로 뽑아 **insertion** 시킴

6가지 punctuation

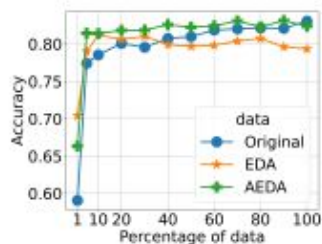
{ " . " , " ; " , " ? " , " : " , " ! " , " , }

4. Experiments

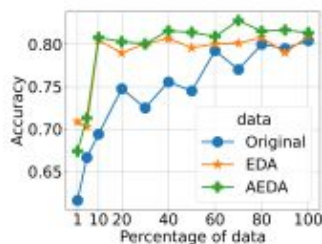
Model	Training set size			
	500	2,000	5,000	full set
RNN	73.5	82.6	85.9	87.9
+EDA	76.1	81.3	85.2	86.5
+AEDA	77.8	83.9	87.2	88.6
CNN	76.5	83.8	87.0	87.9
+EDA	77.5	82.2	84.5	86.1
+AEDA	78.5	84.4	86.5	88.1
Average	75.0	83.2	86.5	87.9
+EDA	76.8	81.8	84.9	86.3
+AEDA	78.2	84.2	86.9	88.4

Table 1: Comparing average performance of EDA and AEDA across all datasets on different training set sizes. For each training sample, 16 augmented sentences were added. Scores are the average of 5 runs.

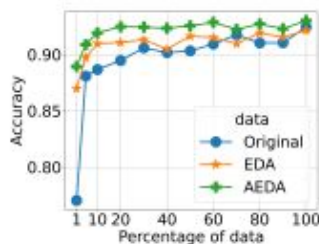
5. Ablation Study



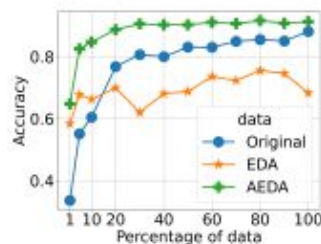
(a) SST-2



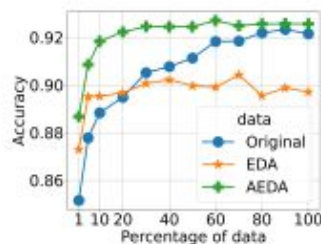
(b) CR



(c) SUBJ



(d) TREC



(e) PC

Figure 2: Performance of the RNN model trained on various proportions of the original, EDA-generated, and AEDA-generated training data for five text classification tasks. All the scores are the average of 5 runs.

5. Ablation Study

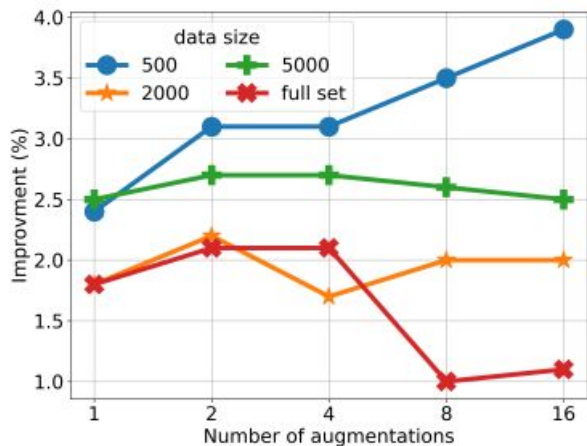


Figure 3: Impact of number of augmentations on the performance of the RNN model trained on various training sizes. Scores are the average of 5 runs over the five datasets. The y axis shows the percentage of improvement.

Model	SST2	TREC
BERT	91.10	97.00
+EDA	90.99	96.00
+AEDA	91.76	97.20

Table 2: Comparing the impact of EDA and AEDA on the BERT model. The model was trained on the combination of the original data and one augmentation for each training sample.

6. Conclusion

1. AEDA를 제안하였다.
(text classification 을 위한 쉬운 data augmentation)
2. 기존의 EDA 기법 보다 간단하지만 , 좋은 성능을 보여주었다.

개인적으로 느낀점.

1. 기존의 NLP Augmentation (EDA , Random delation 같은 것은 오히려 학습에 방해가 된다)
2. BERT 와 같은 Transformer 기반의 실험은 부족
 - a. hidden representation 에서 augmentation 하는 것과 비교 필요
3. 왜 이렇게 좋은 성능 향상을 보였는지 깊게 생각해볼 필요